

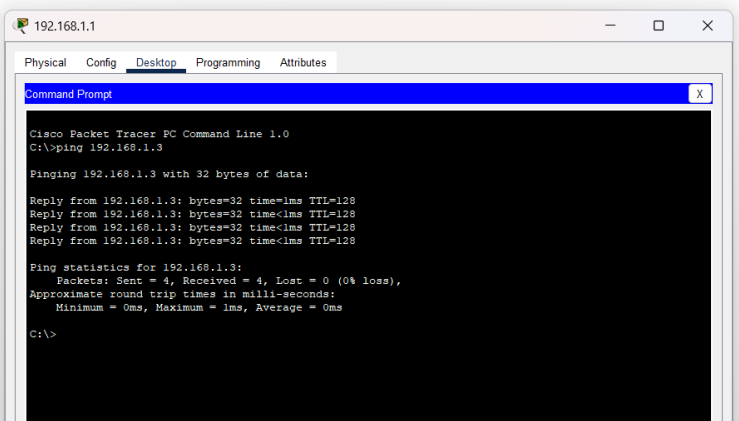
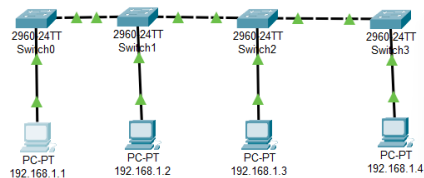
Date: 25/06/2025

Lab Practical #04:

Installation of Network Simulator (Packet Tracer) and Implement different LAN topologies.

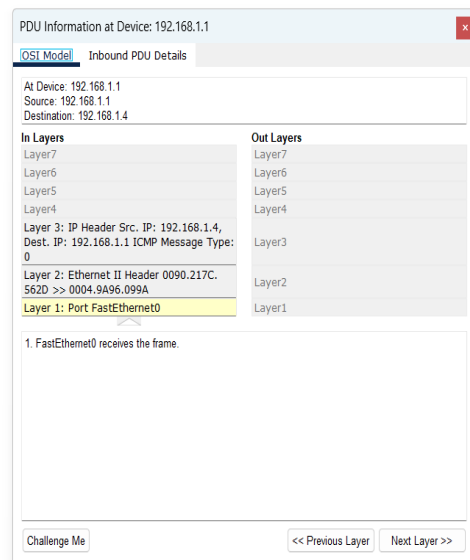
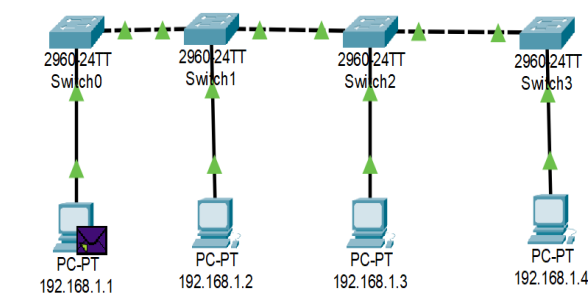
Practical Assignment #04:

1. Create a simple network with switch and two or more pc. Also check connectivity between them using ping command or PDU utility.

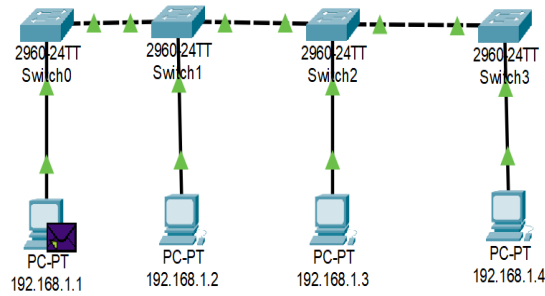


2. Implement different topologies in packet tracer.

a. Bus



Date: 25/06/2025



PDU Information at Device: 192.168.1.1

OSI Model: [Inbound PDU Details](#)

PDU Formats

Ethernet II

PREAMBLE: 101010...10		SF D		DEST ADDR: 0004.9A96.09 9A	
SRC ADDR: 0090.217C.562D	TYPE: 0x0800	DATA (VARIABLE LENGTH)		FCS: 0x00000000	

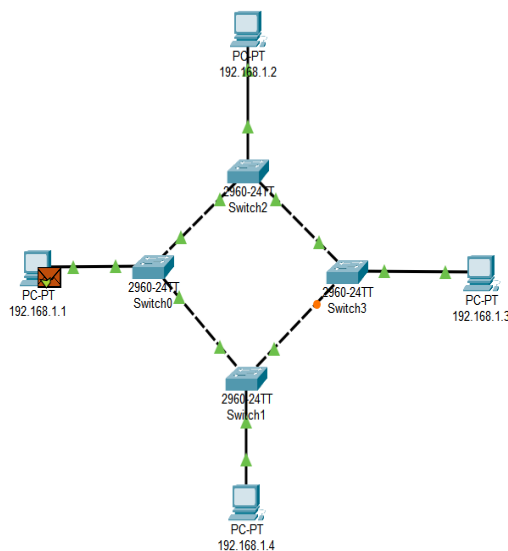
IP

VER: 4	IHL: 5	DSCP: 0x00	TL: 28
ID: 0x0015		FLAGS: 0x0	FRAG OFFSET: 0x000
TTL: 128	PRO: 0x01	CHKSUM	
SRC IP: 192.168.1.4			
DST IP: 192.168.1.1			
DATA (VARIABLE LENGTH)			

ICMP

TYPE: 0x00	CODE: 0x00	CHECKSUM
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b. Ring



PDU Information at Device: 192.168.1.1

OSI Model: [Inbound PDU Details](#)

At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.3

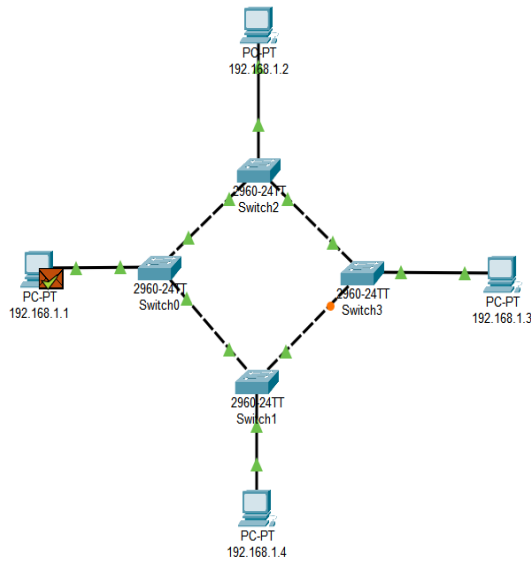
In Layers	Out Layers
Layer 7	Layer 7
Layer 6	Layer 6
Layer 5	Layer 5
Layer 4	Layer 4
Layer 3: IP Header Src. IP: 192.168.1.3, Dest. IP: 192.168.1.1 ICMP Message Type: 0	Layer 3
Layer 2: Ethernet II Header 0001.C701.93AB >> 0090.2B4A.B1BD	Layer 2
Layer 1: Port FastEthernet0	Layer 1

1. FastEthernet0 receives the frame.

Challenge Me

<< Previous Layer Next Layer >>

Date: 25/06/2025



PDU Information at Device: 192.168.1.1

OSI Model [Inbound PDU Details](#)

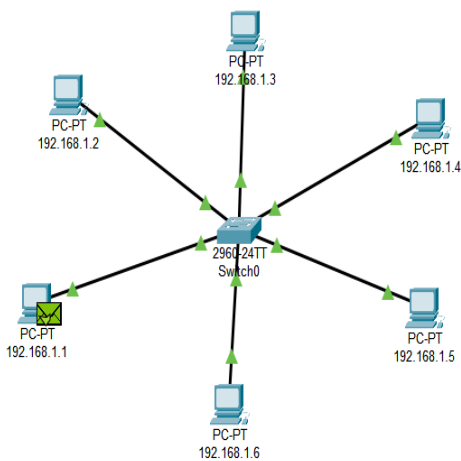
PDU Formats

EthernetII				Bytes
PREAMBLE: 101010...10		SF D	DEST ADDR: 0090.2B4A.B1 BD	
SRC ADDR: 0001.C701.93AB	TYPE: 0x0800	DATA (VARIABLE LENGTH)	FCS: 0x00000000	

IP				Bits
VER: 4	IHL: 5	DSOCP: 0x00	TL: 28	
ID: 0x0002		FLAGS: 0x0	FRAG OFFSET: 0x000	
TTL: 128	PRO: 0x01	CHKSUM		
SRC IP: 192.168.1.3				
DST IP: 192.168.1.1				
DATA (VARIABLE LENGTH)				

ICMP			Bits
TYPE: 0x00	CODE: 0x00	CHECKSUM	

c. Star



PDU Information at Device: 192.168.1.1

OSI Model [Inbound PDU Details](#)

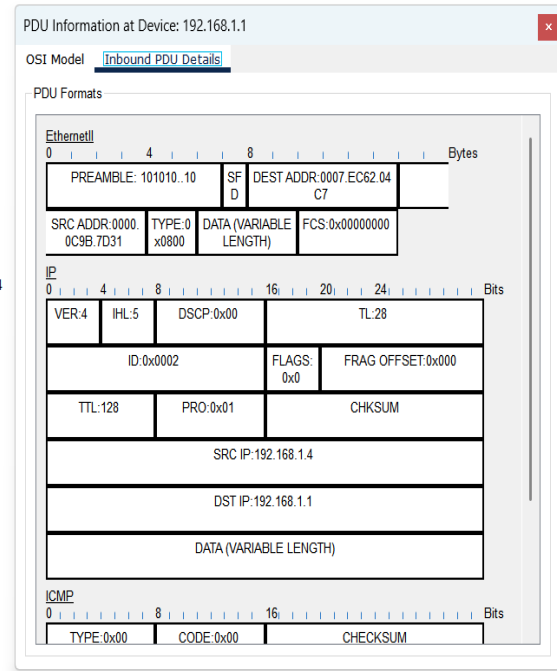
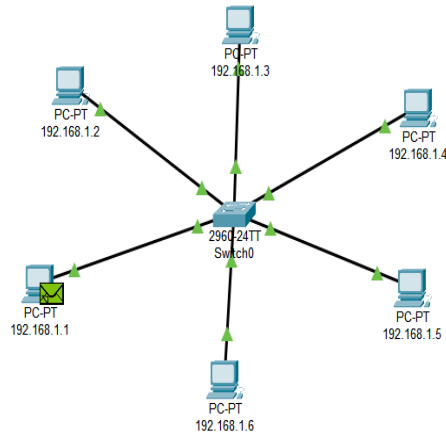
At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.4

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer 3: IP Header Src, IP: 192.168.1.4, Dest, IP: 192.168.1.1 ICMP Message Type: 0	
Layer 2: Ethernet II Header 0000.0C9B.7D31 >> 0007.EC62.04C7	
Layer 1: Port FastEthernet0	Layer1

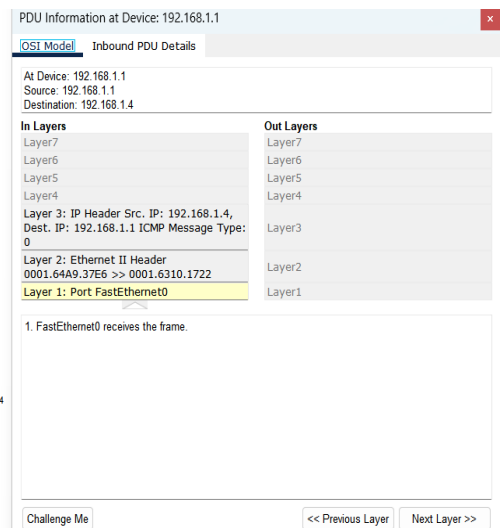
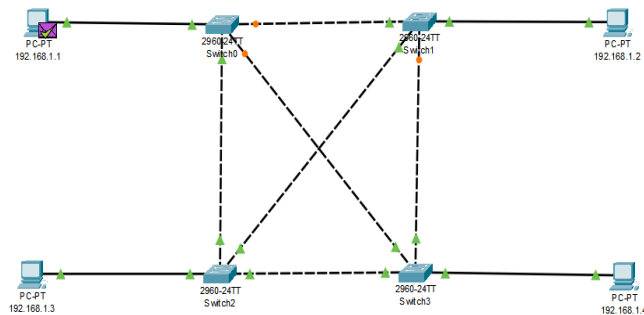
1. FastEthernet0 receives the frame.

Challenge Me [<< Previous Layer](#) [Next Layer >>](#)

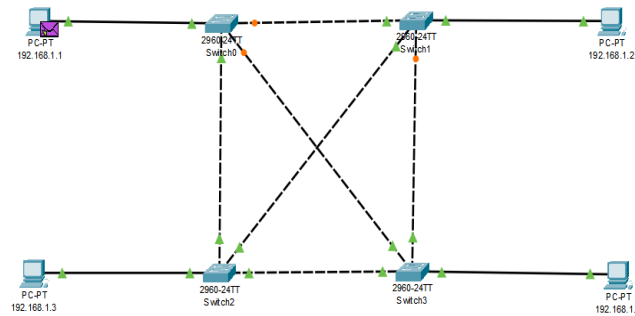
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d. Mesh



Date: 25/06/2025



PDU Information at Device: 192.168.1.1

OSI Model [Inbound PDU Details](#)

PDU Formats

EthernetII

PREAMBLE: 101010...10		SF D		DEST ADDR: 0001.6310.172	
SRC ADDR: 0001.64A9.37E6		TYPE: 0x0800		DATA (VARIABLE LENGTH)	

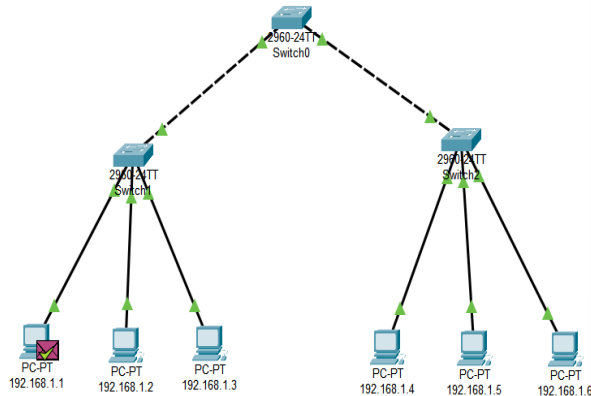
IP

VER: 4	IHL: 5	DSCP: 0x00	TL: 28
ID: 0x0003		FLAGS: 0x0	FRAG OFFSET: 0x000
TTL: 128		PRO: 0x01	CHKSUM
SRC IP: 192.168.1.4			
DST IP: 192.168.1.1			
DATA (VARIABLE LENGTH)			

ICMP

TYPE: 0x00	CODE: 0x00	CHECKSUM
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e. Tree



PDU Information at Device: 192.168.1.1

OSI Model [Inbound PDU Details](#)

At Device: 192.168.1.1
Source: 192.168.1.1
Destination: 192.168.1.6

In Layers

- Layer7
- Layer6
- Layer5
- Layer4
- Layer3: IP Header Src. IP: 192.168.1.6, Dest. IP: 192.168.1.1 ICMP Message Type: 0
- Layer2: Ethernet II Header 00D0.97B0.EA49 >> 00D0.58D9.9A4C
- Layer1: Port FastEthernet0

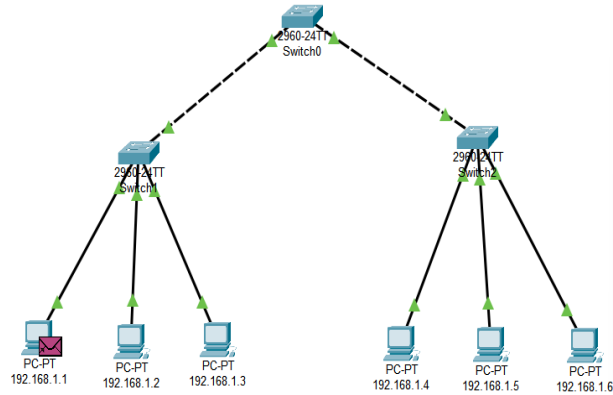
Out Layers

- Layer7
- Layer6
- Layer5
- Layer4
- Layer3
- Layer2
- Layer1

1. FastEthernet0 receives the frame.

Challenge Me << Previous Layer Next Layer >>

Date: 25/06/2025



PDU Information at Device: 192.168.1.1

OSI Model [Inbound PDU Details](#)

PDU Formats

EthernetII									
0 4 8 Bytes									
PREAMBLE: 101010...10						SF D		DEST ADDR: 00D0.58D9.9A4C	
SRC ADDR: 00D0.97B0.EA49		TYPE: 0x0800		DATA (VARIABLE LENGTH)			FCS: 0x00000000		
IP									
0 4 8 16 20 24 Bits									
VER: 4		IHL: 5		DSCP: 0x00			TL: 28		
ID: 0x0002						FLAGS: 0x0		FRAG OFFSET: 0x000	
TTL: 128			PRO: 0x01			CHKSUM			
SRC IP: 192.168.1.6									
DST IP: 192.168.1.1									
DATA (VARIABLE LENGTH)									
ICMP									
0 8 16 Bits									
TYPE: 0x00			CODE: 0x00			CHECKSUM			